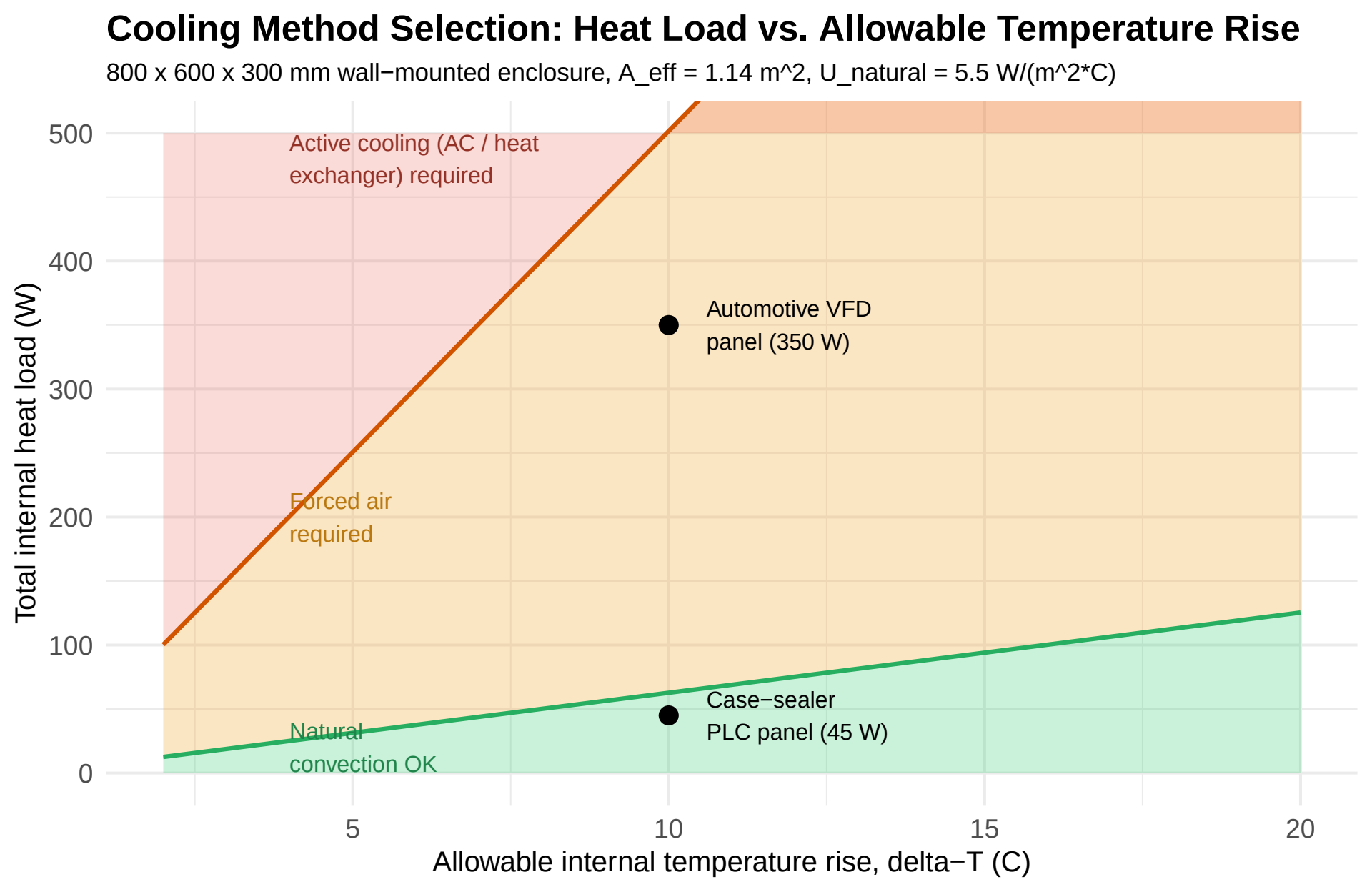
Cooling Method Selection: Heat Load vs. Allowable Temperature Rise
800 x 600 x 300 mm wall−mounted enclosure, A_eff = 1.14 m^2, U_natural = 5.5 W/(m^2*C)
Active cooling (AC / heat exchanger) required
Forced air required
Natural convection OK
Automotive VFD panel (350 W)
Case−sealer PLC panel (45 W)
Total internal heat load (W)
0
100
200
300
400
500
5
10
15
20
Allowable internal temperature rise, delta−T (C)